Smart Lender

AI-Based Loan Eligibility Prediction System Documentation

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Track: AI-ML and Gen AI Track

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1. System Overview

Smart Lender is an AI-powered loan eligibility prediction system that assists banks and financial institutions in evaluating loan applications quickly and accurately. The application integrates a **Machine Learning model** with a **Flask-based web application** and a **MySQL database** to analyze applicant information such as income, employment status, credit score, loan amount, and loan history.

The system predicts whether an applicant is eligible for a loan and provides an approval probability. It minimizes manual verification time, improves consistency in decision-making, and enhances customer experience through a simple web interface.

2. System Architecture

The application follows a standard layered architecture:

- **Presentation Layer** – HTML5, CSS3, JavaScript, Bootstrap
- **Application Layer** – Flask REST API
- **Business Logic Layer** – Authentication, Data Validation, Loan Processing
- **Machine Learning Layer** – Scikit-learn Loan Eligibility Prediction Model
- **Data Layer** – MySQL Database with SQLAlchemy ORMBackend REST API Specifications

POST /api/predict

Receives applicant information, validates input data, predicts loan eligibility using the Machine Learning model, stores the prediction, and returns the eligibility result.

Request Format:-

```
{
  "gender": "Male",
  "married": "Yes",
  "dependents": 1,
  "education": "Graduate",
  "self_employed": "No",
  "income": 55000,
  "coapplicant_income": 12000,
  "loan_amount": 180000,
  "loan_term": 360,
```

"credit_history": 1,
"property_area": "Urban"
}

Response Format:-

{
"loan_status": "Approved",
"approval_probability": "94.5%"
}

3. Setup & Installation Instructions

Follow these steps to set up and run the Vanguard HDI application locally:

- **Step 1 – Create Virtual Environment:-**
  python -m venv venv

- **Step 2 - Activate Virtual Environment:-**
  venv\Scripts\activate

- **Step 3 – Install Dependencies:-**
  pip install -r requirements.txt

- **Step 4 – Configure Database:-**
  Create a MySQL database named:
  smart_lender
  Update database credentials inside
  config.py

- **Step 5 – Train the Machine Learning Model:-**
  python Training/train_model.py
  loan_model.pkl

- **Step 6 – Run the Flask Application:-**
  python app.py

- **Step 7 – Open the Application:-**
  http://127.0.0.1:5000/